



Indian Institute of Information Technology, Surat
Department of Electronics & Communication Engineering
Mid Semester Examination, 3rd Sem, 2025-26
Programming for Problem Solving (CS303)

Timing: 3:30 PM to 5:00 PM

Date: 24/09/2025

Marks:30

Note: Attempt all parts of a question at one place.

1. Logic lock is a two-player game where the second player sets a secret code, and the first player attempts to guess it. In this variation, the secret code is a six-digit number with all distinct digits. After each guess, the second player provides feedback on how many digits in the guessed number are correctly placed in the secret code. For instance, if the secret code is 123456 and the guess is 175286, the score would be two, since the digits 1 and 6 are correctly positioned. (4)

Write an algorithm that, given a sequence of guesses and their respective scores, determines whether there exists a possible secret code that could produce those scores.

Describe following functions with syntax and example: map(), reduce(), filter(), enumerate(). (4)

- (b) Write a python script that allows the user to type in a phrase and then outputs the acronym for that phrase. Note: An acronym is a word formed by taking the first letters of the words in a phrase and making a word from them. For example, RAM is an acronym for 'random access memory'. The acronym should be all uppercase, even if the words in the phrase are not capitalized. (3)

3. (a) Explain Tuples, Lists and Dictionaries with examples and give comments on mutability for each of them. (3 marks)

- (b) Consider the list [9,8,7,6,5,4,3]. Write the Python program which performs the following operation without using built-in methods. (3 marks)

- 1) Insert element 10 at the beginning of the list.
- 2) Insert element 2 at the end of the list.
- 3) Delete the element at index position 5.
- 4) Print all elements in reverse order.

4. (a) What is the lambda function in python? Explain with any (one) example and how to use the lambda function with zip() method with an example. State List comprehension with syntax and use a nested list comprehension to find all of the numbers from 1-1000 that are divisible by any single digit besides 1 (2-9). (3 marks)

5. (a) Short Notes on:
1. Object Oriented Programming
2. Objects and Classes in Python (5 marks)

The word hangman is a simple game. You will provide a word with the first letter. The rest of the letters will be blank. For example, The player has to guess the letters in the word. And the player will guess one letter at a time. If the letter exists in the word, it will appear to the player. If the letter doesn't exist in the word, it will be counted as a wrong try. If the player exceeds the number of the wrong tries, the player will lose. Otherwise, the player will win. Write a Python function to implement this game. (5 marks)

****End****